



This image shows a field of rapeseed that has been genetically modified to be resistant to herbicides. In some countries, as much as 90 percent of all rapeseed is genetically modified.

Mir Space Station

The Soviet space station Mir in orbit about 223 miles (360 km) above the Pacific Ocean. This photograph was taken in 1995, from NASA's space shuttle Discovery.

cosmonauts from 12 different countries, for most of its 15-year lifetime.

Also up in space, five probes flew close to Halley's comet, which was venturing in toward the inner Solar System. The European Space Agency's Giotto probe passed within 373 miles (600 km) of the centre of the comet, capturing the **first ever images of a comet's nucleus**.

Researchers attempting to unravel the **mystery of Alzheimer's disease** discovered the nature of the neurofibrillary tangles (NFTs) found inside neurones in the brains of people with the disease. They found that NFTs are made of Tau (tubule associated unit) proteins, which stabilize microtubules—vital for maintaining cell structure. This was the **second major breakthrough** in understanding Alzheimer's disease since the characterization of beta-amyloid protein (see 1985) that makes up plaques between neurons in Alzheimer's sufferers.

The first field trials of **genetically modified (GM) plants** began in France and the USA: tobacco plants with altered DNA

41.6 MPH THE AVERAGE SPEED OF THE SUNRAYCER SOLAR CAR

that gave them resistance to herbicides. Since 1986, many GM crops have been produced, including cotton, potatoes, and rapeseed. From the outset, the production of **GM crops** has been **controversial**, with concerns about the unpredictable consequences of GM organisms on the environment and some people simply opposed to "tampering" with nature.

In February 1987, astronomers in the Southern Hemisphere observed a **new source of light in the sky**, as bright as many of the stars visible to the naked eye. It turned out to be a supernova—the

Solar car

General Motors' car Sunracer beat the competition in the first ever Solar Challenge, a race for solar powered vehicles across Australia at the height of summer.

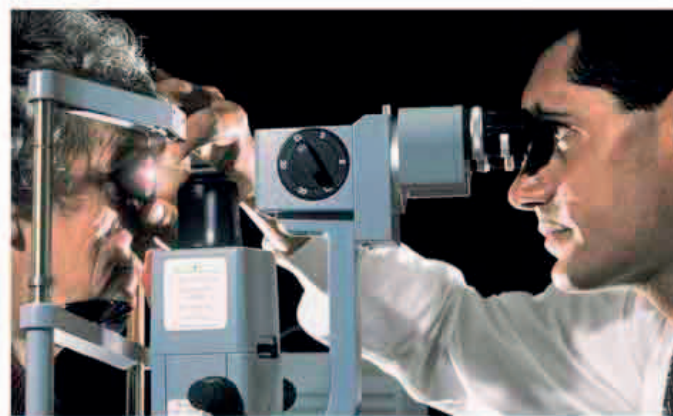
Laser eye surgery

An ophthalmologist prepares to carry out laser eye surgery by first measuring the curvature of the patient's eye.

explosive death throes of a giant star—and was labeled SN 1987a. The explosion happened in a nearby galaxy, the Large Magellanic Cloud, nearly 170,000 light-years away. It was the first naked-eye supernova for more than 300 years.

The discovery of the ozone hole over Antarctica (see 1985) led to more research into the effects of chlorofluorocarbons (CFCs) on the ozone layer. The United Nations proposed an international treaty to **limit the production of CFCs**, and the Montreal Protocol was opened for signature in September 1987, and came into force in 1989.

In Berlin, German ophthalmic surgeon **Theo Seiler** (b.1949) carried out the **first laser eye surgery** on a human patient.



Corrective eye surgery began in the 1970s, with the introduction of radial keratotomy, in which radial lines are cut into the cornea, changing its shape. In 1983, American ophthalmologist **Stephen Trokel** (b.1934) developed a way of **changing the cornea's shape** by burning away corneal tissue with an ultraviolet laser. It was this method, called photo-refractive keratectomy

(PRK), that Seiler was using. A more sophisticated procedure, called LASIK (laser in-situ keratomileusis) was patented in 1989 and available commercially from 1991.

In November, the first World Solar Challenge—a race held to promote research into **solar technology for cars**—was held in Australia. The race was won by General Motors' Sunracer.



February 23, 1987 Supernova SN1987a, bright enough to be observed with the naked eye

March 1987 The first licensed AIDS treatment, AZT, is approved in the US

March 1987 Scientists observe that brain cells in Alzheimer's patients have defective Tau protein

October 1987 Chinese-born American bioengineer Bert Fung coins the term "tissue engineering"

November 1987 Sunracer wins the first ever World Solar Challenge for solar-powered cars

November 1987 World's first laser eye surgery is performed on a human being